1. Create C/Yacc code to generate 3-address code for a set of given valid C statements. This is to be done by implementing SDT with any parsing technique.

The input will be a .C file. The 3-address code needs to be generated for all arithmetic and logical statements present in the C file. Any other statements need to be ignored.

Some Examples

a = b + c * d	100: t1 = c * d 101: t2 = b + t1 102: a = t2
x = (a + b) * (c - d)	100: t1 = a + b 101: t2 = c - d 102: t3 = t1 * t2 103: x = t3
x = (a < b)	100: if a < b goto 103 101: t1 = 0 102: goto 104 103: t1 = 1 104: x = t1
if (a < b) x = 1;	100: if a < b goto 102 101: goto 103 102: x = 1

A sample input and output file

<table>
<tr>
<td>#include <stdio.h>

int main() {
   float balance = 1000.50;
   float rate = 0.05;
   int years = 5;
   int is_premium_member = 1;
   int account_active = 1;

   float interest = balance * (1 + rate) * years - balance;
   balance = balance + interest;

   if ((balance > 1200.00 && is_premium_member) || account_active ==</td>
<td>100: balance = 1000.50
101: rate = 0.05
102: years = 5
103: is_premium_member = 1
104: account_active = 1
105: t1 = 1 + rate
106: t2 = balance * t1
107: t3 = t2 * years
108: t4 = t3 - balance
109: interest = t4
110: balance = balance + interest
111: if balance > 1200.00 goto 113
112: goto 114
113: if is_premium_member != 0 goto 116
114: if account_active == 1 goto 116</td>
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```
1) {
        balance += 50.0;
        years++;
    }

    if (!(balance < 500.00) && years != 0) {
        balance = balance - 10.0;
    }

    return 0;
}
```

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115: goto 120
116: t5 = balance + 50.0
117: balance = t5
118: t6 = years + 1
119: years = t6
120: if balance < 500.00 goto 124
121: if years == 0 goto 124
122: t7 = balance - 10.0
123: balance = t7
124: return
```